SAMREEN KAUR

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GitHub: https://github.com/samreen1410 | Portfolio: https://samreen1410.github.io/portfolio/

Summary

Fourth-year Science student at the University of British Columbia with hands-on experience in software development, application testing, API validation, database analysis, and data analytics through professional and academic projects. Proficient in Python, Java, SQL, C++, and R, with experience using Oracle SQL, Azure, OpenShift and GitHub Actions in technical environments. Demonstrated ability to analyze complex systems, validate data accuracy, debug automated tests, and build data-driven solutions. Strong problem-solving, communication, and teamwork skills with experience working in collaborative technical environments. Passionate about building practical, reliable software that solves real problems for the people who use it. Seeking software engineering, QA automation, or data-focused roles where I can contribute from day one and keep growing as an engineer.

Skills

- **Programming:** Python, Java, C++, C, SQL, R, JavaScript, HTML, CSS
- **Software & Development Tools:** Git, Jira, Confluence, IntelliJ IDEA, Jupyter, RStudio, Linux, Windows, macOS
- **Cloud & Development Platforms:** Azure, AWS, IBM Cloud, Google Cloud, GitHub Actions, OpenShift (OCP)
- **Data & Analytics:** Oracle SQL, Sybase, Apache Spark, TensorFlow, Pandas, NumPy, MATLAB, Data Analysis, Statistical Analysis, Machine Learning, Deep Learning, Data Visualization
- **Technical Skills:** Data Structures & Algorithms, Software Testing, API Testing, Regression Testing, Database Validation, Batch Processing, Debugging
- **Interpersonal:** Problem Solving, Critical Thinking, Teamwork, Time Management, Written & Verbal Communication

Education

- University of British Columbia, Vancouver, BC — Expected graduation: December 2026
  Bachelor of Science

Professional Experience

Software/Application Developer Co-op - CIBC — May 2026 – Aug 2026
Toronto, ON

- Performed production and system validation for Private Wealth applications by reconciling API/XML responses and account-level totals against expected results, identifying discrepancies and data anomalies across DIT and SIT testing cycles.
- Executed 20+ batch testing scenarios across multiple environments using Git, Azure, OpenShift, and GitHub Actions, generating input files and validating outputs.
- Debugged Java-based automated test scripts and executed API regression test cases during an ESB modernization SIT certification cycle, validating services supporting account information, trade confirmations, tax receipts, and report retrieval.
- Wrote and executed Oracle SQL queries to verify backend database updates, confirm successful record creation, and reconcile application results during application and batch testing.
- Created and tracked Jira tickets for defects, test scenarios, and validation activities, while documenting technical workflows and batch processes in Confluence.

Data Analyst – Data Science UBC — May 2023 – Aug 2023
Vancouver, BC

- Collected and analyzed data from SQL databases to support development of a sustainable campus project.
- Used Python and R for data processing, analysis, and development of a working analytical design.
- Collaborated with a data science research team to investigate and communicate data-driven findings.

Technical Projects

Metro Vancouver Housing Affordability Tool Aug 2026 – Present

- Built and deployed a full stack web app that estimates realistic living costs across Metro Vancouver cities and projects when independent living becomes affordable for the user.
- Developed a Python backend hosted on Render that caches CMHC and Statistics Canada data, keeping responses fast and within free tier limits.
- Created a rent explorer for comparing rents city by city and an interactive affordability calculator, with frontend deployed on GitHub Pages.

Exoplanet Transit Signal Classification Jan 2026 – Apr 2026

- Trained and compared five machine learning models (Logistic Regression, kNN, Decision Trees, Random Forest, SVM) to classify exoplanet transit signals from NASA TESS data.
- Addressed severe class imbalance using precision, recall, and F1-score, and optimized a Random Forest via grid search with cross-validation.
- Result: achieved 93.9% accuracy and 0.95 AUC, with transit depth, orbital period, and duration identified as the most predictive features.

Interactive Data Visualization Website Sept 2025 – Dec 2025

- Designed and developed an interactive data visualization website to analyze demographic patterns in billionaire wealth accumulation using a large, multi-source dataset.
- Developed linked and parameterized visualizations (scatterplots, heatmaps, density and violin plots, etc) using Altair (Python) to support exploratory analysis.

Traffic Accident Project May 2025 – Aug 2025

- Analyzed 10 years of open-sourced traffic accident data using Apache Spark.
- Trained machine learning models to classify accident severity; achieved 22% higher F1-score than baseline.
- Deployed model with Google Cloud and dashboard for real-time scenario testing.

Used Car Selling Price Analysis Sept 2024 – Jan 2025

- Fitted a regression model and used exhaustive selection model to identify covariates with a significant effect on used car selling prices.
- Result: The variables associated with selling price of a used car are the present price, the year, the number of previous owners, the seller type and the mileage of the vehicle.

International Student Proportion Predictive Model Jan 2024 – May 2024

- Tested whether university ranking variables can predict the proportion of international students, using forward selection to build an RMSE-reduced model.
- Correlation analysis showed the model performed best with four input variables.

Certifications

- Specialization
  HTML, CSS, and JavaScript for Web Developers - Johns Hopkins
- Professional Certificate
  IBM Data Science - Advanced Python

Additional Experience

Tutor high school students – Coding and Mathematics Jan 2023 – Aug 2023
Orientation leader – Science UBC Aug 2023 – Oct 2023
Customer Service Associate – Save on Foods Jul 2023 – May 2025
Science Undergraduate Society UBC – Associate leader Oct 2022